

Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

OBSERVATION:

1. GPIO ports of AT89C51 were studied and their pins were configured as input/output.
2. Bit-addressable memory locations and individual port bits were accessed and controlled independently.
3. Bit addressing allows specific bits to be set, cleared, or toggled without affecting other bits.

PROGRAM:

```
#include <reg51.h>

sbit LED1 = P1^0;
sbit LED2 = P1^1;

void delay()
{
    unsigned int i,j;
    for(i=0;i<200;i++)
    for(j=0;j<1000;j++);
}

void main()
```

```

{
while(1)
{
LED1 = 1;
LED2 = 0;

delay();

LED1 = 0;
LED2 = 1;

delay();

}

}

```

OUTPUT:

